Ex: $\eta_T = 0,5 \rightarrow$ microstrip antenna efficiency

$G_T = 11dB \rightarrow$ antenna gain

$P_{in} = 40W$ (input power before antenna)

Tx power density fall as $\left(\frac{1}{d^2}\right)$

1) $P_{in} = _____ ??$ [dBm]

$$P_{in_{dBm}} = 10\log\left(\frac{P_{in}}{10^{-3}}\right)^{40} = 46dBm$$

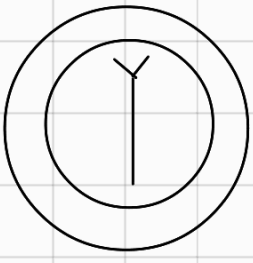
$$P_{T\,dBm} = P_{in\,dBm} - \underbrace{\eta_{T\,dB}}_{\underbrace{10\log(0,5)}_{3dB}}$$

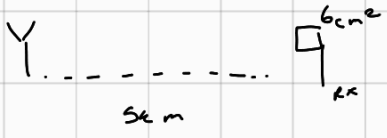
2) $P_T = ?$ [dBm]

$$P_T = P_{in} \cdot \eta_T = 40 \cdot 0,5 = 20W$$

$$P_{T_{dBm}} = 10\log\left(\frac{20}{10^{-3}}\right) = 43dBm$$

c) $d = 5km \rightarrow P_D = ?$





$$P_D = \frac{P_T \cdot G_T}{4\pi d^2}$$

$$P_D = \frac{20 \cdot 10^{11/10}}{4\pi (5 \cdot 10^3)^2} = 1,603 \mu W/m^2$$

d) $A_R = 6cm^2 \rightarrow P_R = ?$

$$P_R = P_D \cdot A_R = 1.603 \frac{\mu W}{m^2} \cdot (6cm^2)$$

$$= 1,603 \frac{\mu W}{m^2} (6 \cdot 10^{-4}) m^2$$

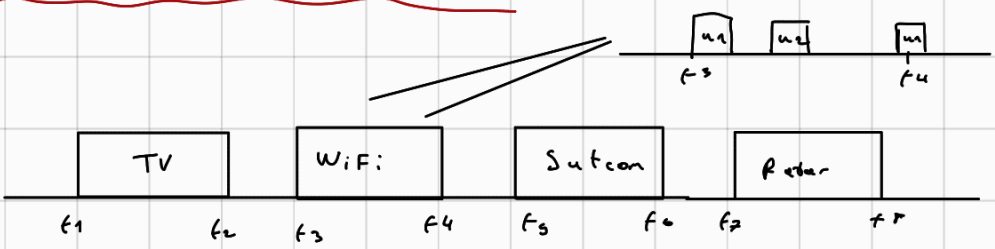
$$= P_R = 962pW$$

e) Alicidak gürültü

$N_R = 100fW \rightarrow SNR_R = ?$

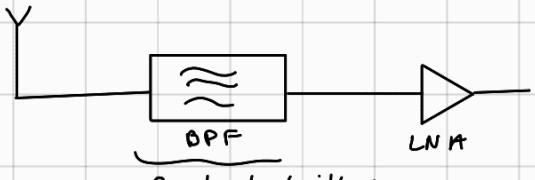
$$SNR = \frac{P_R}{N_R} = \frac{962pW}{100fW} = 9618 \rightarrow SNR_{dB} = 39,8dB$$

Design Architecture of Receivers

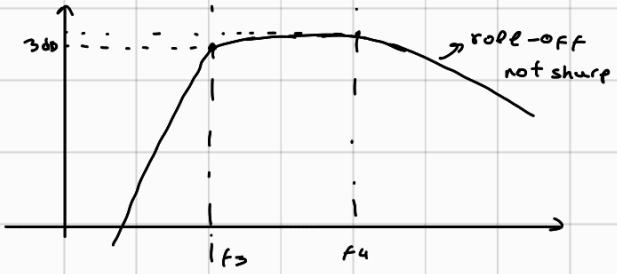


Band → frequency allocated for specific application

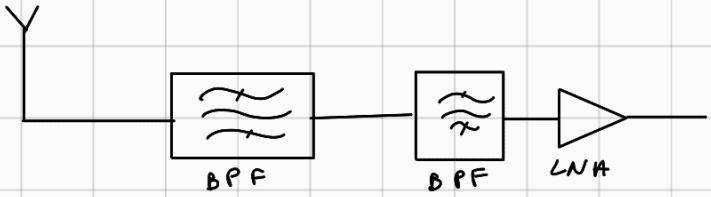
Channel → " " " " user



Band select filter



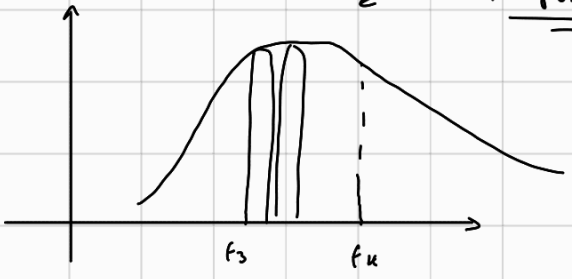
off-chip filter → eğer filtrenin keskin olmasını istersek



Band select

Tunable channel select

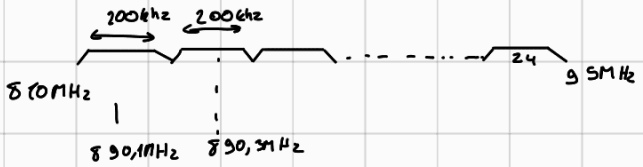
not possible



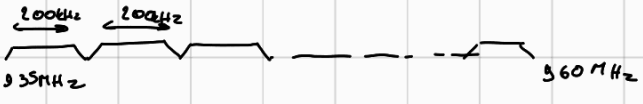
Narrowband sharp-roll-off filter is not possible

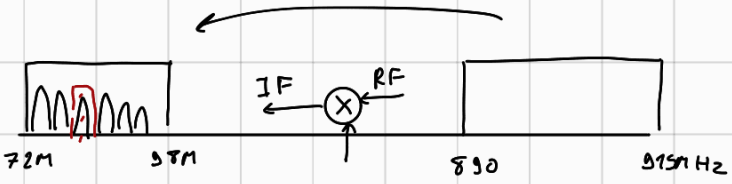
GSM 900 band

Uplink band & channels



Downlink band & channels

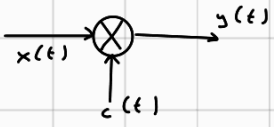




Down Conversion

$x(t) = A\cos(\omega_{RF} t)$ input

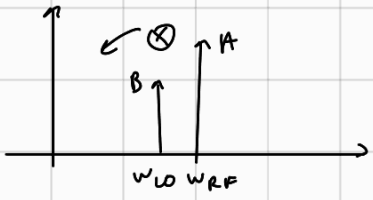
$c(t) = B\cos(\omega_{LO} t)$ LO

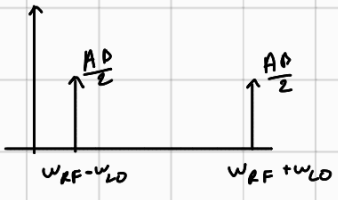


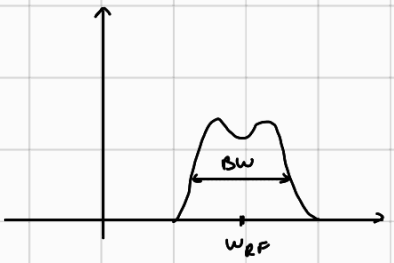
$$r(t) = y(t) = x(t)\,c(t) = A\cos(\omega_{RF} t)\; B\cos(\omega_{LO} t)$$

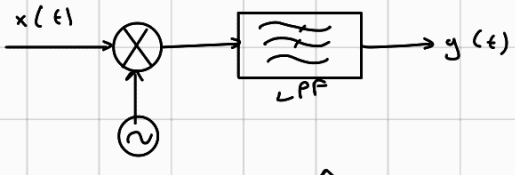
$\omega_{RF} > \omega_{LO}$

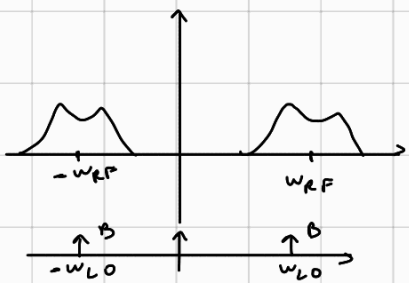
$$= \frac{AB}{2}\left\{\cos(\omega_{RF}+\omega_{LO})t + \cos(\omega_{RF}-\omega_{LO})t\right\}$$

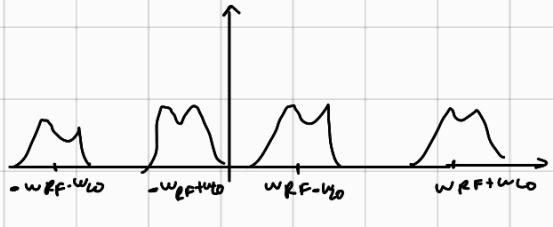




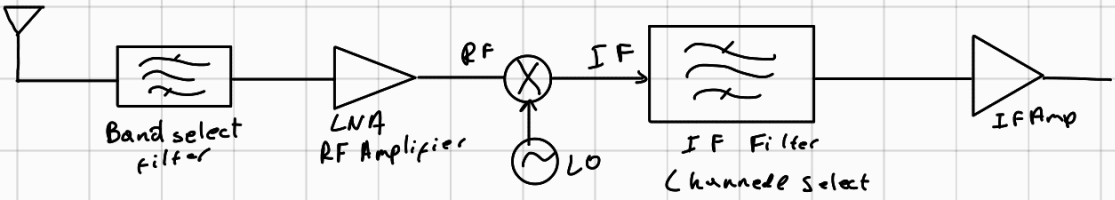






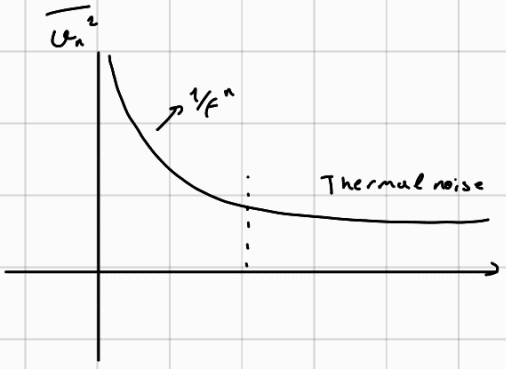


Heterodyne Receivers



$\omega_{RF} - \omega_{LO} = \omega_{IF} \neq 0 \rightarrow$ Heterodyne

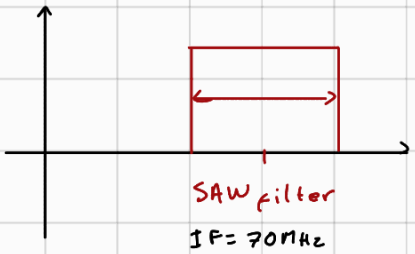
$\omega_{IF} = 0 \rightarrow$ Homodyne





1- Tunable IF Filter, constant LO

2- Constant IF Filter, Tunable LO



IF = 70MHz

BW = 100MHz

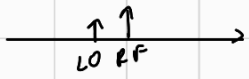
ω_{RF}	ω_{LO}	ω_{IF}
↓	↓	constant
960	890	70
965	895	70

High side injection

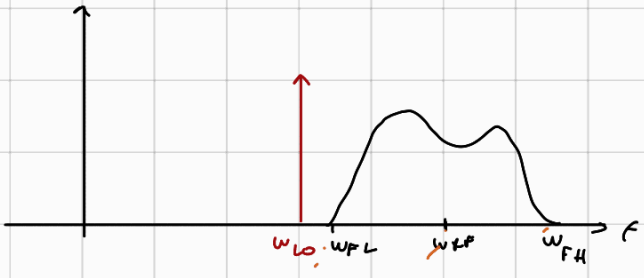


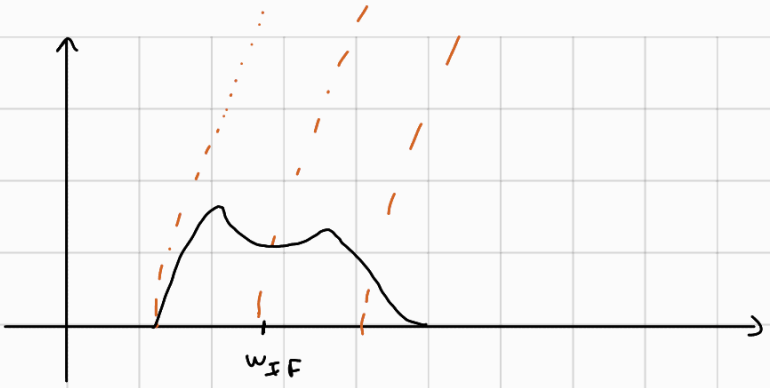
$\rightarrow \omega_{IF} = \omega_{LO} - \omega_{RF}$

Low side injection



$\rightarrow \omega_{IF} = \omega_{RF} - \omega_{LO}$





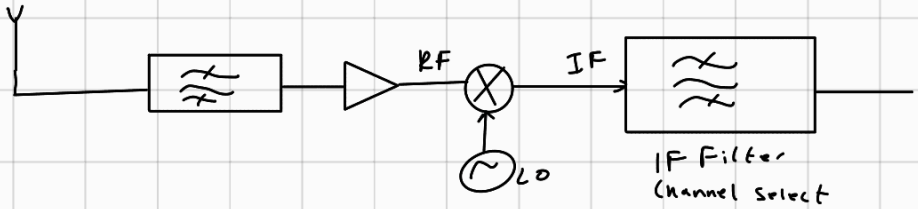


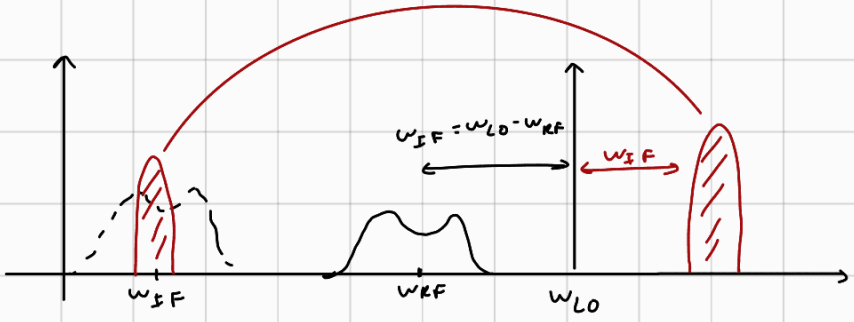
Spectrum Inversion



High-side Injection

Image Frequency: Hayal Frekans



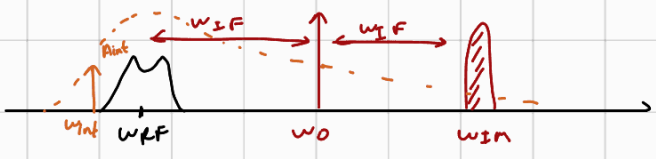


$\omega_{IM} - \omega_{LO} = \omega_{IF}$

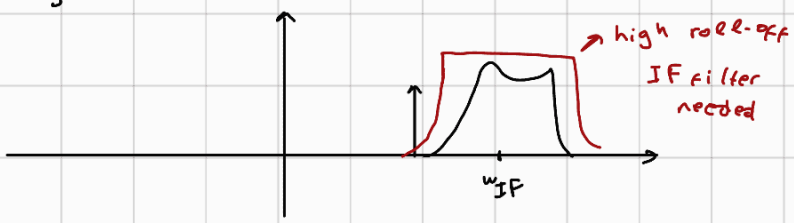
$\omega_{LO} - \omega_{RF} = \omega_{IF}$

2-s cenarios

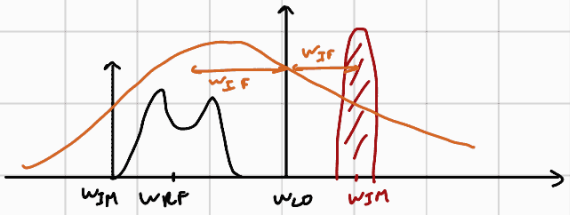
1) ω_{LO} far from ω_{RF} and high ω_{IF}



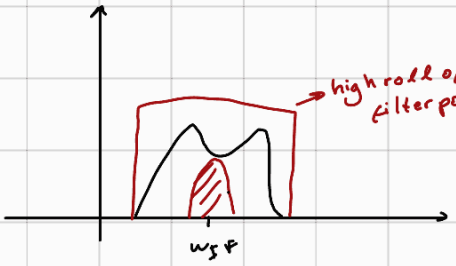
IF bölgesi



2) ω_{LO} close to ω_{RF} and low ω_{IF}



$\omega_{IF_2} < \omega_{IF_1}$



$IP_{3\,IFAmp_2} > IP_{3\,IFAmp_1} > IP_{3\,LNA}$

ilk çevrimde uzak bir LO kullanabilirsemokadar filtreden faydalanır

$\omega_{IF_1} > \omega_{IF_2}$

Dual Conversion Receiver

Noise critical

Image Rejected Filter

$\omega_{IF} >> \omega_{IF_1} = \omega_{LO} - \omega_{RF}$

Linearity critical

$\omega_{LO_2} - \omega_{IF_1} = \omega_{IF_2}$

baseferi yakından karşılamalı

